

Set 2: Customer Feedback Analysis

Experiment 1: Histogram of Customer Ages

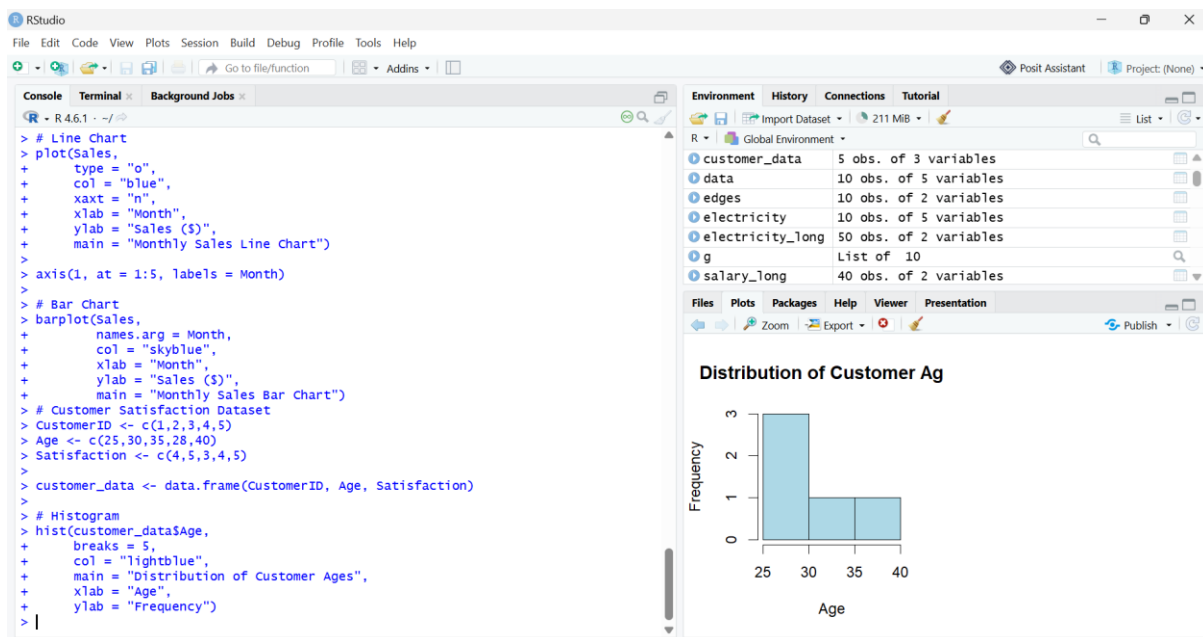
```
CustomerID <- c(1,2,3,4,5)
```

```
Age <- c(25,30,35,28,40)
```

```
Satisfaction <- c(4,5,3,4,5)
```

```
customer_data <- data.frame(CustomerID, Age, Satisfaction)
```

```
hist(customer_data$Age,  
      breaks = 5,  
      col = "lightblue",  
      main = "Distribution of Customer Ages",  
      xlab = "Age",  
      ylab = "Frequency")
```



Experiment 2: Pie Chart of Customer Satisfaction Scores

```
CustomerID <- c(1,2,3,4,5)
```

```
Age <- c(25,30,35,28,40)
```

```
Satisfaction <- c(4,5,3,4,5)
```

```
customer_data <- data.frame(CustomerID, Age, Satisfaction)
```

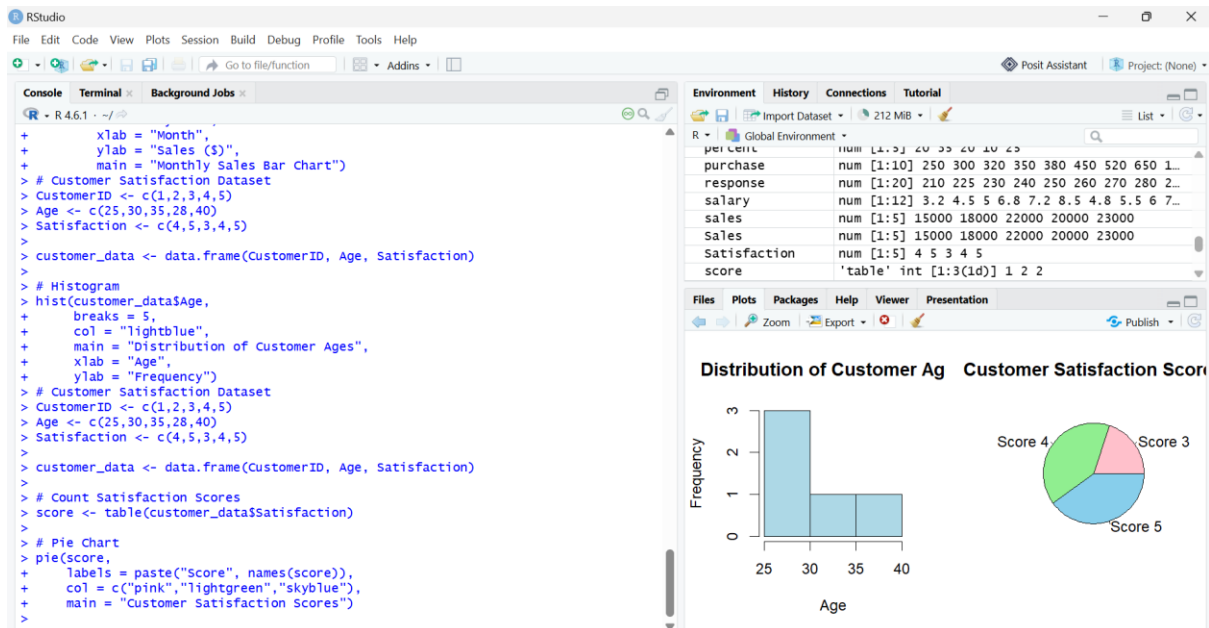
```
score <- table(customer_data$Satisfaction)
```

```
pie(score,
```

```

labels = paste("Score", names(score)),
col = c("pink","lightgreen","skyblue"),
main = "Customer Satisfaction Scores")

```



Experiment 3: Stacked Bar Chart

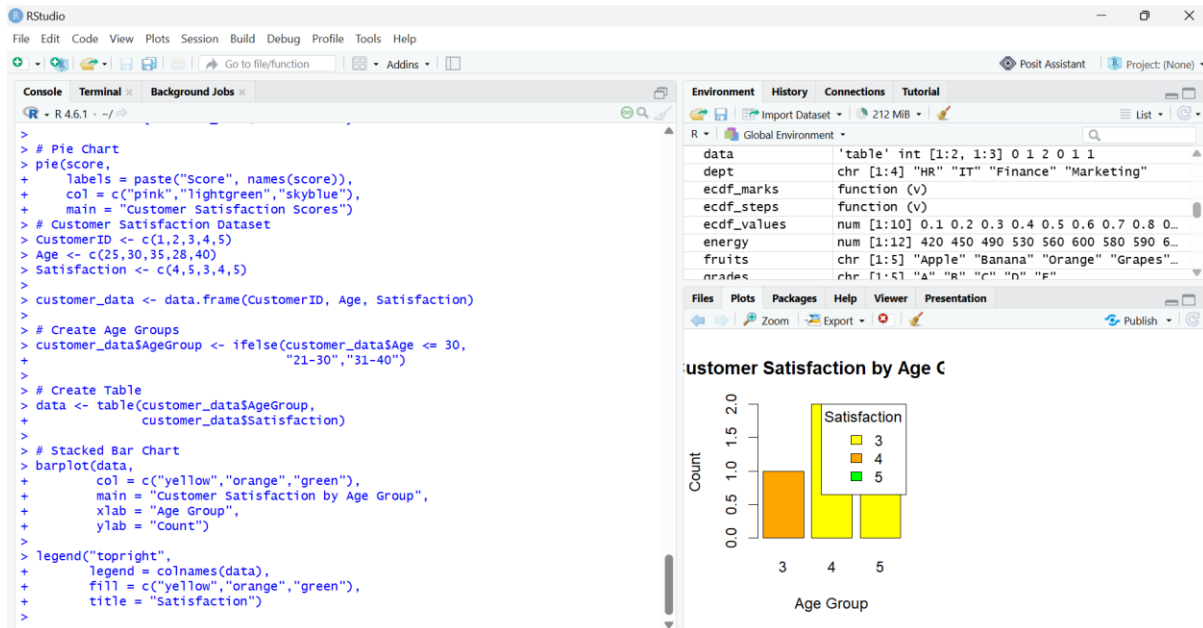
```

CustomerID <- c(1,2,3,4,5)
Age <- c(25,30,35,28,40)
Satisfaction <- c(4,5,3,4,5)
customer_data <- data.frame(CustomerID, Age, Satisfaction)
customer_data$AgeGroup <- ifelse(customer_data$Age <= 30,
                                "21-30","31-40")
data <- table(customer_data$AgeGroup,
              customer_data$Satisfaction)
barplot(data,
        col = c("yellow","orange","green"),
        main = "Customer Satisfaction by Age Group",
        xlab = "Age Group",
        ylab = "Count")
legend("topright",
      legend = colnames(data),

```

```
fill = c("yellow","orange","green"),

title = "Satisfaction")
```



Experiment 4: Word Cloud

```
install.packages("tm")

install.packages("wordcloud")

install.packages("RColorBrewer")

library(tm)

library(wordcloud)

library(RColorBrewer)

feedback <- c(

  "Good service and fast delivery",

  "Excellent product quality",

  "Very good customer support",

  "Fast delivery and excellent service",

  "Good experience and quality product")

docs <- Corpus(VectorSource(feedback))

docs <- tm_map(docs, content_transformer(tolower))

docs <- tm_map(docs, removePunctuation)

docs <- tm_map(docs, removeNumbers)
```

```
docs <- tm_map(docs, removeWords, stopwords("english"))
```

```
tdm <- TermDocumentMatrix(docs)
```

```
m <- as.matrix(tdm)
```

```
word_freq <- sort(rowSums(m), decreasing = TRUE)
```

```
wordcloud(
```

```
  words = names(word_freq),
```

```
  freq = word_freq,
```

```
  min.freq = 1,
```

```
  random.order = FALSE,
```

```
  colors = brewer.pal(8, "Dark2"))
```

